CARD 01

# Propane

IUPAC name: propane

LEVEL: FOUNDATION

$C_3H_8$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$C_3H_8$

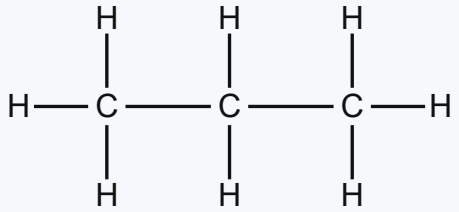
Given together with the compound name; reproduced exactly

| Atoms | C | H | Total | match |
|---|---|---|---|---|
| In the molecular formula | 3 | 8 | **11** | |
| Counted in the Lewis structure | 3 | 8 | **11** | ✓ |

Valence electrons: 20 available = 20 in 10 bonds + 0 in 0 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE




No lone pairs: every valence electron is in a bond. Every atom is shown; no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

Each carbon is written together with the hydrogens attached to it.

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 02

# Ethanol

IUPAC name: ethanol

LEVEL: FOUNDATION

$C_2H_6O$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_2H_6O$$

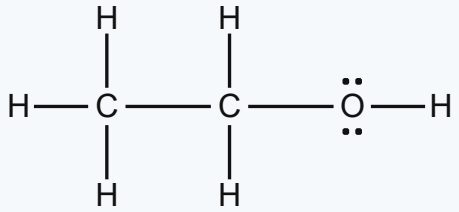
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 2 | 6 | 1 | **9** | |
| Counted in the Lewis structure | 2 | 6 | 1 | **9** | ✓ |

Valence electrons: 20 available = 16 in 8 bonds + 4 in 2 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



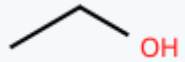

Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

Each carbon is written with its hydrogens; the hydroxyl group is kept as OH.

## 4. BOND-LINE FORMULA (KETCHER)




Drawn in Ketcher.

CARD 03

# Acetone

IUPAC name: propan-2-one

LEVEL: FOUNDATION

$C_3H_6O$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_3H_6O$$

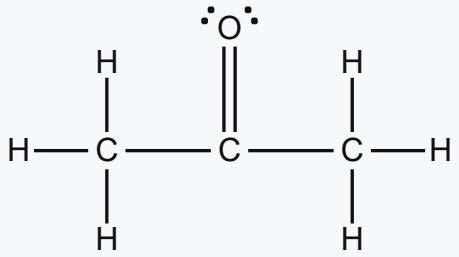
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 3 | 6 | 1 | **10** | |
| Counted in the Lewis structure | 3 | 6 | 1 | **10** | ✓ |

Valence electrons: 24 available = 20 in 10 bonds + 4 in 2 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



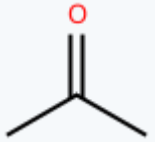

Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)CH_3$$

(=O) is an oxygen double-bonded to the carbon written just before it (the carbonyl carbon).

## 4. BOND-LINE FORMULA (KETCHER)




Drawn in Ketcher.

CARD 04

# Acetic acid

IUPAC name: ethanoic acid

LEVEL: DEVELOPING

$C_2H_4O_2$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_2H_4O_2$$

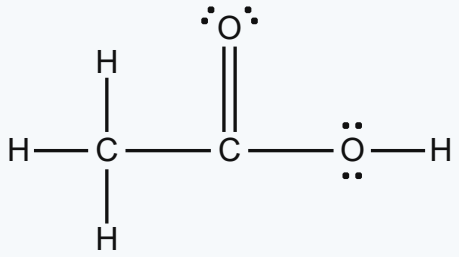
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 2 | 4 | 2 | **8** | |
| Counted in the Lewis structure | 2 | 4 | 2 | **8** | ✓ |

Valence electrons: 24 available = 16 in 8 bonds + 8 in 4 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



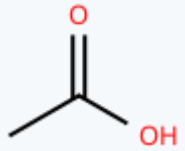

Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)OH$$

(=O) is the carbonyl oxygen; the final OH is the hydroxyl of the carboxylic acid group.

## 4. BOND-LINE FORMULA (KETCHER)




Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 05

# Ethyl acetate

IUPAC name: ethyl ethanoate

LEVEL: DEVELOPING

$C_4H_8O_2$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_4H_8O_2$$

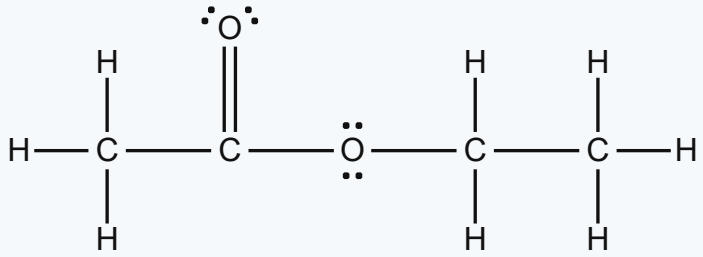
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 4 | 8 | 2 | **14** | |
| Counted in the Lewis structure | 4 | 8 | 2 | **14** | ✓ |

Valence electrons: 36 available = 28 in 14 bonds + 8 in 4 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



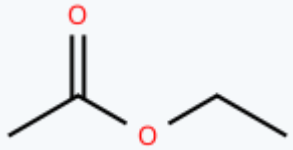
Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)OCH_2CH_3$$

(=O) is the carbonyl oxygen; the next O is the ester oxygen joining the acetyl carbon to $CH_2CH_3$.

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 06

# Cyclohexane

IUPAC name: cyclohexane

LEVEL: DEVELOPING

$C_6H_{12}$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_6H_{12}$$

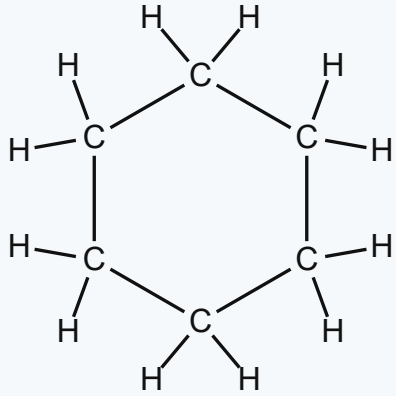
Given together with the compound name; reproduced exactly

| Atoms | C | H | Total | match |
|---|---|---|---|---|
| In the molecular formula | 6 | 12 | **18** | |
| Counted in the Lewis structure | 6 | 12 | **18** | ✓ |

Valence electrons: 36 available = 36 in 18 bonds + 0 in 0 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE




No lone pairs: every valence electron is in a bond. Every atom is shown; no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

cyclo-$(CH_2)_6$

Six $CH_2$ groups joined end to end in a closed ring: every $CH_2$ is bonded to two neighbouring $CH_2$ groups.

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 07

# Cyclohexanone

IUPAC name: cyclohexanone

LEVEL: DEVELOPING

$C_6H_{10}O$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_6H_{10}O$$

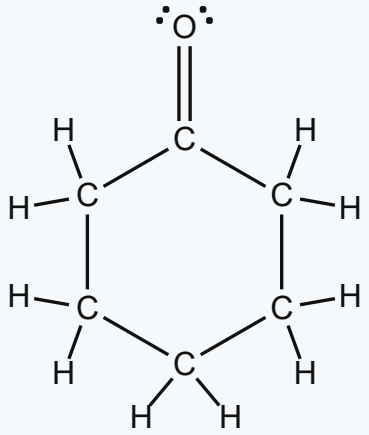
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 6 | 10 | 1 | **17** | |
| Counted in the Lewis structure | 6 | 10 | 1 | **17** | ✓ |

Valence electrons: 40 available = 36 in 18 bonds + 4 in 2 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



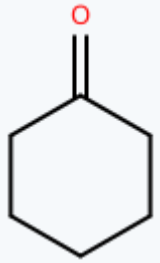

Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

cyclo-$(CH_2)_5C{=}O$

Five $CH_2$ groups and one C=O carbon joined in a closed six-membered ring.

## 4. BOND-LINE FORMULA (KETCHER)




Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 08

# Acetaminophen

also paracetamol | IUPAC name: N-(4-hydroxyphenyl)acetamide

LEVEL: CHALLENGE

$C_8H_9NO_2$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_8H_9NO_2$$

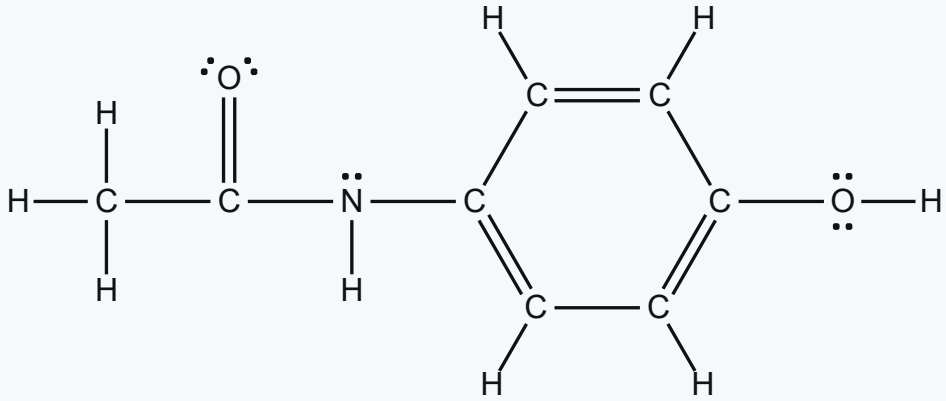
Given together with the compound name; reproduced exactly

| Atoms | C | H | N | O | Total | match |
|---|---|---|---|---|---|---|
| In the molecular formula | 8 | 9 | 1 | 2 | **20** | |
| Counted in the Lewis structure | 8 | 9 | 1 | 2 | **20** | ✓ |

Valence electrons: 58 available = 48 in 24 bonds + 10 in 5 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



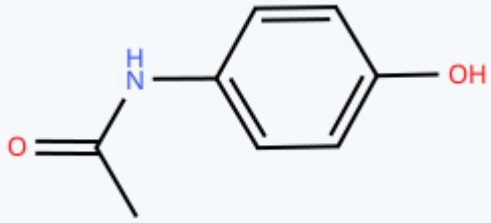
Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)NH–C_6H_4–OH$$

$–C_6H_4–$ is a benzene ring carrying two substituents on opposite carbons (para, C-1 and C-4): $NHC(=O)CH_3$ on C-1 and OH on C-4.

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 09

# Aspirin

also acetylsalicylic acid | IUPAC name: 2-(acetyloxy)benzoic acid

LEVEL: CHALLENGE

$C_9H_8O_4$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_9H_8O_4$$

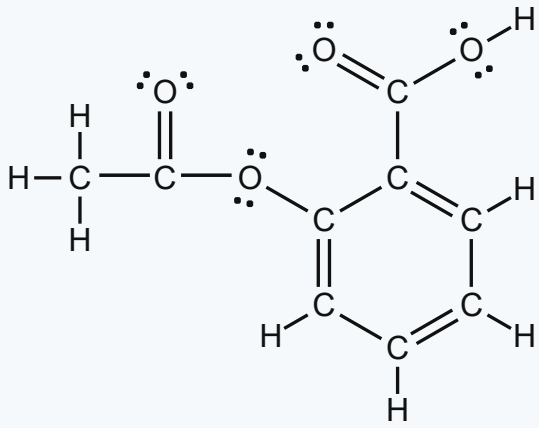
Given together with the compound name; reproduced exactly

| Atoms | C | H | O | Total | match |
|---|---|---|---|---|---|
| In the molecular formula | 9 | 8 | 4 | **21** | |
| Counted in the Lewis structure | 9 | 8 | 4 | **21** | ✓ |

Valence electrons: 68 available = 52 in 26 bonds + 16 in 8 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



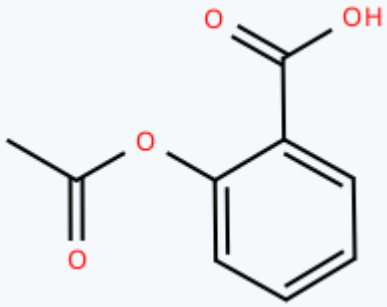
Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)O–C_6H_4–C(=O)OH$$

$–C_6H_4–$ is a benzene ring carrying two substituents on adjacent carbons (ortho, C-1 and C-2): $OC(=O)CH_3$ (acetoxy) on C-1 and $C(=O)OH$ (carboxylic acid) on C-2.

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity

CARD 10

# Caffeine

IUPAC name: 1,3,7-trimethyl-3,7-dihydro-1H-purine-2,6-dione

LEVEL: CHALLENGE

$C_8H_{10}N_4O_2$

## 1. MOLECULAR FORMULA (identity and atom-count check)

$$C_8H_{10}N_4O_2$$

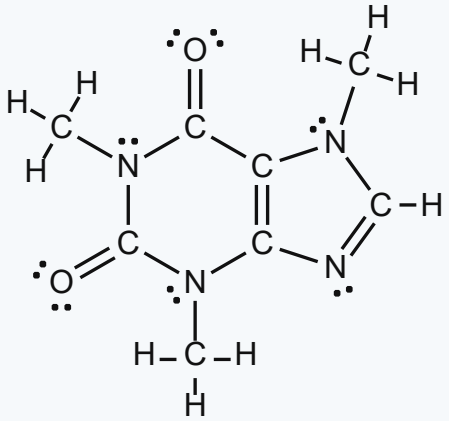
Given together with the compound name; reproduced exactly

| Atoms | C | H | N | O | Total | match |
|---|---|---|---|---|---|---|
| In the molecular formula | 8 | 10 | 4 | 2 | **24** | |
| Counted in the Lewis structure | 8 | 10 | 4 | 2 | **24** | ✓ |

Valence electrons: 74 available = 58 in 29 bonds + 16 in 8 lone pairs

Octet on every C, N, O; duet on every H; no formal charges needed

## 2. LEWIS STRUCTURE



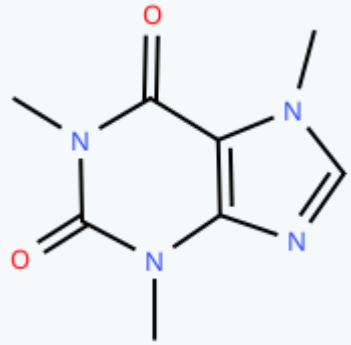
Dots = lone pairs. Every atom is shown. All atoms are neutral: no formal charges required.

## 3. CONDENSED STRUCTURAL FORMULA

$CH_3–N^1–CH=N–C^2=C^1–C(=O)–N(CH_3)–C(=O)–N^2–CH_3$

Fused rings are written as one path. The superscript digits mark ring closures: the two atoms carrying the same digit are bonded to each other ($N^1–C^1$ and $C^2–N^2$).

## 4. BOND-LINE FORMULA (KETCHER)



Drawn in Ketcher.

☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity